Understanding SVM Kernels: A Comparative Study of Linear, Polynomial, and RBF Kernels

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Github: https://github.com/Suryaaaaa9999/svm-kernel-tutorial.git

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1. Introduction :

Support Vector Machines (SVMs) are a powerful class of supervised machine learning algorithms used for classification and regression tasks. Their primary objective is to find an optimal decision boundary (also known as a hyperplane) that separates data points belonging to different classes while maximizing the margin between them.

In many real-world scenarios, data is not linearly separable. To address this, SVMs use a technique known as the **kernel trick**, which transforms the input data into a higher-dimensional space where a linear separation becomes possible.

This tutorial focuses on understanding how different kernel functions influence the behavior and performance of SVM models. Specifically, we compare three commonly used kernels:

- Linear Kernel
- Polynomial Kernel
- Radial Basis Function (RBF) Kernel

The aim is to demonstrate how kernel choice affects classification accuracy and decision boundaries, enabling better model selection in practice.

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2. Dataset Description :

The dataset used in this study is the **Pima Indians Diabetes Dataset**, a well-known medical dataset used for binary classification tasks.

Dataset Overview

- Total samples: **768**
- Features: **8 numerical input features**
- Target variable: **Binary (0 = No Diabetes, 1 = Diabetes)**

The dataset used in this tutorial is a binary classification dataset with two numerical features:

- **Feature 1**
- **Feature 2**

The goal is to classify observations into two classes. The dataset exhibits non-linear patterns, making it suitable for comparing kernel-based methods.

Preprocessing Steps :

Data split into:

- **Training set (70%)**
- **Test set (30%)**

Before training, feature scaling is applied using **StandardScaler**, which standardizes features to have zero mean and unit variance. This is crucial for SVM performance since SVM is sensitive to feature scaling.
The dataset is slightly imbalanced but still suitable for classification analysis.

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3. Methodology :

3.1 Model Pipeline

A machine learning pipeline is used to ensure consistent preprocessing and modeling:

1. **StandardScaler** → Normalizes features
2. **SVC (Support Vector Classifier)** → Performs classification
   This prevents data leakage and ensures fair evaluation.

3.2 Kernel Functions

Linear Kernel

- Produces a straight decision boundary
- Works well when data is linearly separable
- Fast and computationally efficient

Polynomial Kernel

- Produces curved decision boundaries
- Controlled by the **degree** parameter
- Can model feature interactions

RBF Kernel

- Highly flexible and non-linear
- Uses distance-based similarity
- Controlled by **gamma** parameter
- Often best for complex datasets

3.3 Hyperparameter Tuning

To optimize performance, **GridSearchCV** with **5-fold cross-validation** was used.

Parameters tuned:

- **C** → Controls margin vs misclassification trade-off
- **degree** → Polynomial complexity
- **gamma** → Influence of training points
  This ensures the best parameter combination is selected for each kernel.

4.Results :

Kernel Train Accuracy Test Accuracy :

Polynomial - 0.75 , 0.736
RBF - 0.79 , 0.758
Linear - 0.78 , 0.749

Figure 1: Polynomial kernel decision boundary showing curved separation. Performance depends on degree selection.

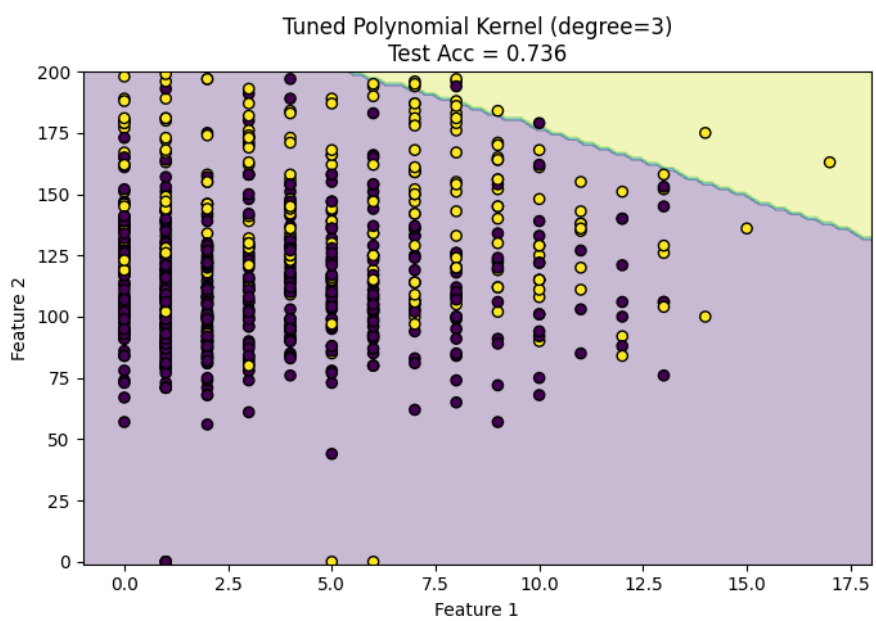


Figure 2: Linear kernel decision boundary forming a straight line, suitable for simpler data patterns

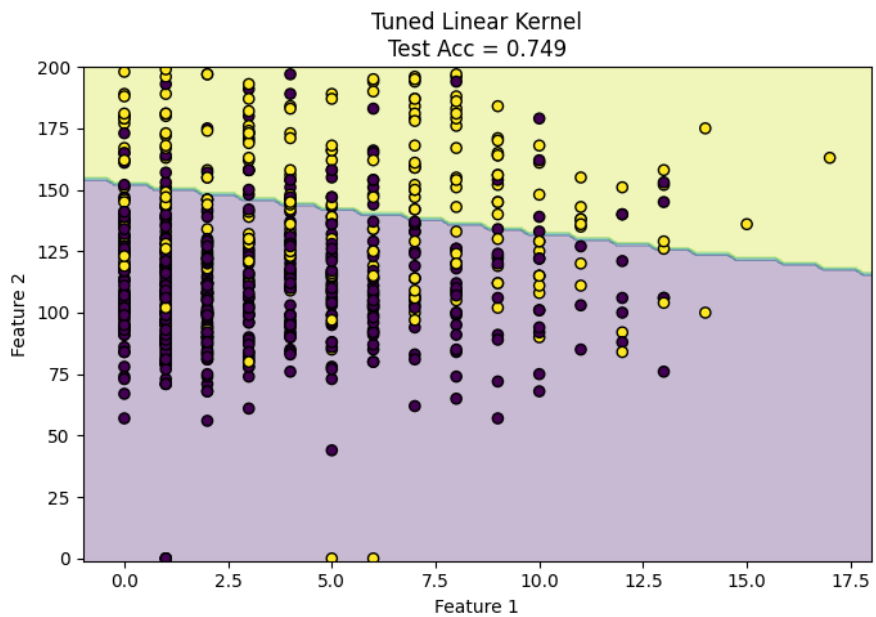


Figure 3: RBF kernel decision boundary showing flexible adaptation to complex data patterns

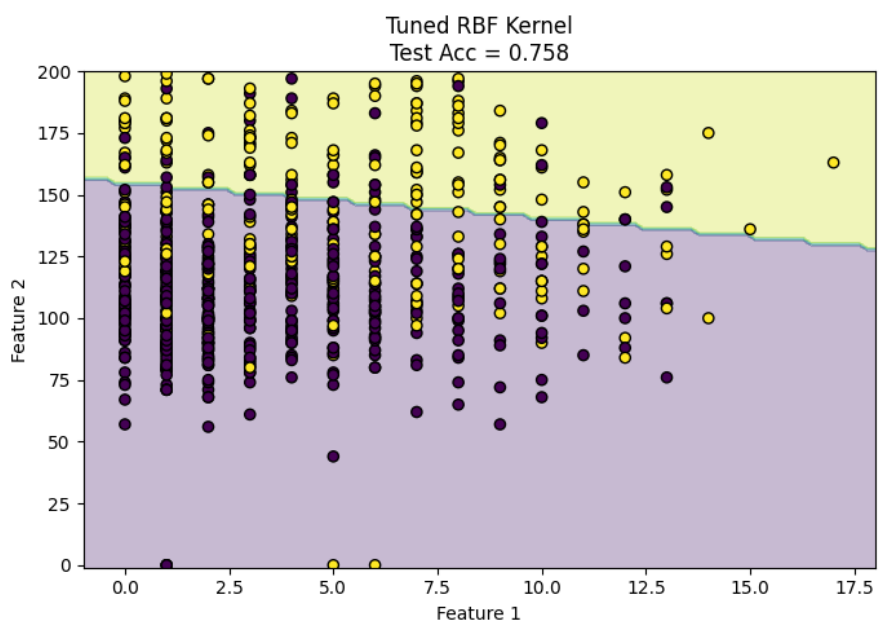
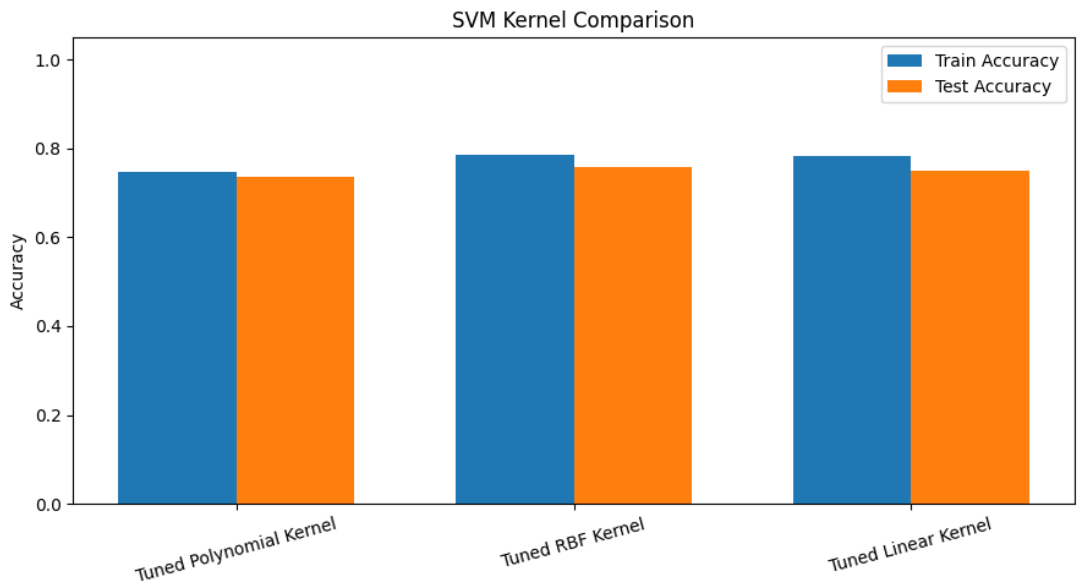


Figure 4: Comparison of training and test accuracy across SVM kernels. RBF shows the best generalization performance.



As shown in Figure 3, the RBF kernel creates a more flexible decision boundary.

The performance of the three SVM kernels—Polynomial, Linear, and RBF—was evaluated using both training and test accuracy. The results show that the **RBF kernel achieved the highest test accuracy of approximately 0.758**, indicating the best generalization performance among the three models. It also recorded the highest training accuracy (around 0.79), suggesting that it effectively captures the underlying patterns in the data without significant overfitting.

The **linear kernel performed slightly lower**, with a test accuracy of approximately 0.749 and a training accuracy of about 0.78. This indicates that while the dataset contains some linear structure, a purely linear decision boundary is not sufficient to fully capture the complexity of the data.

The **polynomial kernel achieved the lowest performance**, with a test accuracy of approximately 0.736 and a training accuracy of around 0.75. This may be due to the sensitivity of the polynomial kernel to the degree parameter, which can lead to underfitting or overfitting depending on its value.

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5. Discussion :

5.1 Why did RBF perform best?

The RBF kernel maps data into a high-dimensional space and considers distances between points. This allows it to adapt to complex patterns and local variations in the dataset, leading to higher accuracy.

5.2 Why did the Polynomial Kernel underperform?

The polynomial kernel underperformed compared to the other kernels primarily because it is highly sensitive to the choice of the degree parameter. A lower degree can result in underfitting, where the model is too

simple to capture the underlying patterns in the data. On the other hand, a higher degree can lead to overfitting, where the model becomes overly complex and fits noise rather than the true structure. In this case, the selected degree may not have been optimal for representing the dataset complexity, which likely contributed to its lower performance compared to the RBF and linear kernels.

5.3 Insights from the Linear Kernel

The linear kernel performed reasonably well, indicating that the dataset contains some degree of linear separability. Its relatively good performance suggests that a simple decision boundary can capture a portion of the underlying structure in the data. Additionally, linear models are computationally efficient and easier to interpret, making them a strong baseline choice. However, because the dataset includes non-linear relationships, the linear kernel lacks the flexibility needed to achieve the highest possible accuracy, which explains why it performs slightly worse than the RBF kernel.

5.4 Overfitting Analysis

The comparison between training and test accuracy across all three models shows only a small difference, indicating that overfitting is minimal. This suggests that the models generalize well to unseen data and are not excessively memorizing the training set. The use of cross-validation during hyperparameter tuning also contributes to this stability, as it ensures that the selected models perform consistently across different subsets of the data. Overall, the results demonstrate a good balance between bias and variance for all kernels, with the RBF kernel achieving the best trade-off.

The results highlight the bias-variance tradeoff, where linear models have higher bias and RBF models achieve lower bias at the cost of increased flexibility.

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6. Conclusion :

This study demonstrates that the choice of kernel has a significant impact on the performance of Support Vector Machine models. In medical applications such as diabetes prediction, capturing non-linear relationships is critical for improving diagnostic accuracy.

The results show that the **RBF kernel provides the best overall performance**, as it is able to effectively model the non-linear relationships present in the dataset. Its flexibility allows it to adapt to complex patterns, leading to higher accuracy and better generalization. The **linear kernel**, while simpler, still performs reasonably well, suggesting that some degree of linear separability exists within the data. However, its limited flexibility prevents it from achieving the highest performance. The **polynomial kernel**, on the other hand, underperforms due to its sensitivity to the degree parameter, which can make it either too simple or too complex depending on the setting.

Overall, this analysis highlights that **no single kernel is universally optimal**, and the choice of kernel should depend on the structure of the dataset and the problem being solved. In practical applications, the RBF kernel is often a strong default choice for non-linear data, while linear models may be preferred for simpler or more interpretable solutions. Understanding the strengths and limitations of each kernel is essential for building effective and reliable machine learning models

7. References :

1. Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning.
2. Towards Data Science. Support Vector Machines Explained.